Faster, Higher, Further Apart

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Abstract

In light of recent failures by the German Athletics Association (DLV) at the international level, this report examines the structural development of the extended national elite. Using a dataset of over 226,000 entries from the German rankings from 2001 to 2025, we analyze the impact of the COVID-19 pandemic and long-term shifts in performance spreads. We utilize World Athletics Scores to make performances comparable across disciplines. Our results show a significant decline in performance during the pandemic (-2.14% median) and an increasing polarization of the national elite. While the absolute top performers improved, the extended elite declined. This decrease in performance density is particularly evident among men and in technical disciplines and suggests structural deficiencies in coaching presence and internal competition.

1. Introduction

As the world's largest athletics association, the Deutscher Leichtathletik-Verband (DLV) is tasked with facilitating peak individual performance and maintaining an internationally competitive national team (Deutscher Leichtathletik-Verband, n.d.). However, zero medals at the 2023 World Athletics Championships in Budapest and a historic low medal count at the 2024 Paris Olympics have left the organization in a sobering position. Although individual athletes still occasionally reach the podium, the declining return on investment in terms of Olympic success (Fremerey et al., 2024) raises questions about the underlying health of the national talent pool.

While medals are the most visible metric of success, they reflect only the peak of elite performance and offer little insight into how performance is distributed within the elite level. To examine potential structural shifts, we analyze the top 35 rankings from 2001 to 2025 based on two research questions. First, we exploratively assess the impact of the COVID-19 pandemic as a short-term disruption to the performance development. Second, motivated by observed performance drop-offs beyond the top positions, we assess whether the performance gap within the elite has widened over time. Such a trend may reflect reduced internal competitive intensity, which can weaken international success, as elite development relies on strong competitive environments at the club level (Tihanyi, 2001, cited in Green & Houlihan, 2005).

Our analytical approach is detailed in Section 2, where we describe the data processing and justify the analytical framework. Building on this, we will analyze performance metrics to identify structural trends and evaluate our research questions in Section 3. Finally, Section 4 discusses the implications of these findings.

2. Data and Methods

2.1. Data Acquisition and Standardization

The data were retrieved from the official archive of the DLV, documenting top performances by German athletes in national and international competitions across major athletic disciplines (running, hurdles and steeplechase, jumping, throwing, and combined events) from 2001 to 2025. Performances are classified by sex (male, female) and age group (U14 to open class).

Data were extracted from various PDF files using regular expressions to handle formatting variations across years. Outliers and entry inconsistencies (e.g., irregular time formats) were corrected via external sources where possible. Minor data gaps from parsing errors were manually completed by referencing the original files. Non-recurrent distances (e.g., 7.5 km youth events) were excluded, resulting in a final dataset of 238,187 entries.

We standardized discipline names, birth years and converted performance marks into numeric values (seconds for track events and meters for field events) to allow statistical analyses. To enable comparisons across disciplines, all performance marks were transformed into World Athletics scor-

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ing points (IAAF scores), which map heterogeneous results onto a common scale (Wor). As no official formula for score computation is publicly available, we implemented reverse-engineered, discipline-specific quadratic functions based on 2025 scoring tables (Chen, 2019). The following analyses are based on these standardized IAAF scores.

2.2. Data Selection and Grouping

To ensure a representative sample of elite performance, the analyses include the top 35 national results per year, sex, and age group. This threshold establishes a truncated distribution, focusing specifically on the high-performance tail of athletic performances. A "Group" is defined as the combination of sex, age category, and discipline, consistent with official leaderboards. Although the dataset covers a broader range of age categories, the analyses focus on the U18, U20, and Adult groups, as these are the primary age groups competing at international championship level. To facilitate cross-disciplinary comparison, disciplines were aggregated into four primary clusters based on current team selection criteria (DLV): sprints, runs, throws, and jumps. Due to regulatory changes in hurdle heights and throwing weights, affected disciplines were excluded from median and rank-specific analyses but retained for gap evaluations to maintain longitudinal comparability.

2.3. Analysis Plan: Impact of COVID-19

To investigate the impact of the COVID-19 pandemic and to get an overall impression of the data, an exploratory analysis was conducted. A heatmap displaying the number of competitive events per month and year was used to examine seasonal competition patterns and identify disruptions in event frequency and data density during the pandemic period.

To isolate the effect of the pandemic year, performances were grouped into two cohorts: a pre-pandemic baseline comprising pooled IAAF scores from 2015–2019, and a pandemic cohort consisting of all IAAF scores recorded in 2020. General performance shifts were assessed using the relative percentage difference between period medians. Changes in performance density were evaluated using the interquartile range (IQR). The median and IQR were selected as robust measures of central tendency and dispersion, as our data are non-normally distributed and may include extreme values. The primary indicator of distributional change was the relative percentage change in median and IQR between 2020 and the baseline period.

Applicable to the whole study, a positive percentage change in the median represents performance improvement, while a positive change in IQR indicates a widening performance gap. We determined the statistical significance of median shifts using the Mann-Whitney U test. To evaluate changes in performance density, we utilized bootstrap resampling (B=10,000), which is particularly robust for our specific sample sizes ([Was hier gemeint? Vorschlag Gemini:] N ranging from [X] to [Y] per group). Significant changes in density were identified when the 95% confidence intervals (CIs) for the variance differences excluded zero.

2.4. Analysis Plan: Widening Performance Gap

To analyze long-term structural changes, we compare the five-year periods 2001–2005 and 2021–2025. This timeframe smooths out annual fluctuations and, through a broader data base, ensures the statistical validity of the results. The analysis is conducted on two levels: 1. Recording performance trends and density for all groups to identify group differences. 2. Examining the development to identify the structure in which performance changes occur. This allows us to identify general trends and determine the structure of the changes.

To measure the performance density, we define the performance gap (G_p) between the three best and the three worst (e.g., 33rd–35th place) in our data set for the two time periods. The averaging of the top 3 reflects the international standard of success (podium), while the bottom 3 provide a stable and comparable baseline. We used the performance gap to measure the breadth of performance at the top level and utilized all available information (e.g., rankings) from our dataset to measure polarization. Mathematically, the gap (G_p) is defined as follows:

$$G_p = \left(\frac{1}{3|P|} \sum_{y \in P} \sum_{r=1}^{3} x_{y,r} \right) - \left(\frac{1}{3|P|} \sum_{y \in P} \sum_{r=33}^{35} x_{y,r} \right) \tag{1}$$

where in our case $|P| = 5$.

where $x_{y,r}$ represents the individual performance of rank r in year y, and $|P|$, in our case $|P| = 5$, represents the number of years per period. The development of the performance gap was calculated as the relaive percentage change between the two five-year periods.

To analyze the development of the different rankings, we calculate the arithmetic mean for each rank across all groups for the start and end periods. For each rank, even when differentiating by age group and gender, the sample size is at least n = 80. According to the central limit theorem, we can assume a normal distribution and thus calculate the arithmetic mean. The development of each rank was determined by calculating the percentage change between the start and end periods, normalized to the start period. A positive value indicates an improvement in the average ranking, a negative value a decline.

During the period under review, several rule changes were implemented regarding hurdle heights and throwing weights, particularly in the U18 (2012) and U20 age groups (2001) (Pro). Therefore, the affected disciplines were excluded from median and rank analyses, as changes in these parameters are directly attributable to equipment changes and not to athletic development. However, the performance gap analysis was retained, as this variance can be calculated independently of the absolute values.

3. Results

Analysis of the distribution of seasonal events reveals a significant temporal shift in the competition calendar. In 2020, events were concentrated in the latter part of the year, which marked a clear departure from the traditional midsummer peak. This shift toward late summer and autumn remains partially visible in 2021, though the schedule shows a gradual return to the pre-pandemic seasonal norm.

The Mann-Whitney-U test confirms a significant decline in performance levels across all athletic categories during the 2020 season compared to the 2015–2019 baseline ($p < 0.05$). The overall median IAAF score decreased by -2.14%, falling from 935 to 915 points. Throws saw the largest decline (-2.84%), followed by Jumps (-2.45%) and Sprints (-1.94%), while Runs remained most resilient (-0.85%).

The bootstrap analysis of the Interquartile Range (IQR) reveals significant shifts in performance dispersion during the 2020 season compared to the 2015–2019 baseline. Bootstrap analysis revealed significant increases in performance dispersion for Sprint ($+12.04\%$) and Run ($+6.41\%$), while the other disciplines Throw, Jump and remained statistically stable.

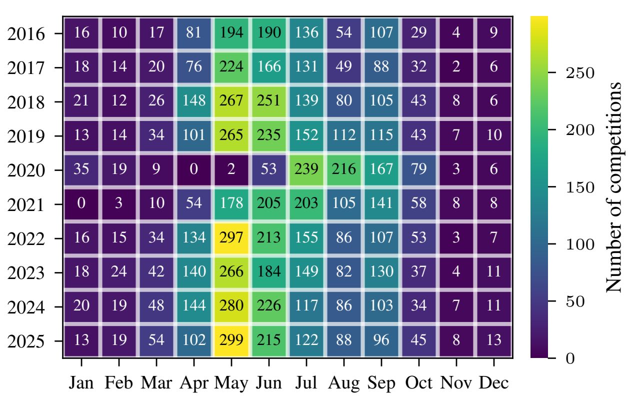

Figure 1.

Figure 3 illustrates a long-term performance shift across 106 groups between 2001-2005 and 2021-2025. While me-

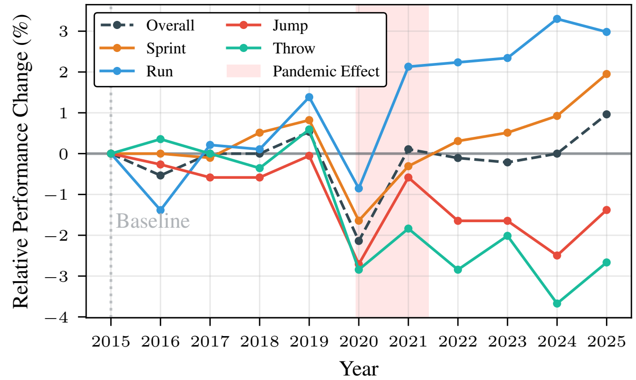

Figure 2.

dian trends were nearly balanced (38 improvements vs. 37 declines), performance density decreased significantly: the performance gap widened in 49.06% of groups ($n = 52$) and narrowed in only 4.72% ($n = 6$). Running and sprinting were the top performers, with improvements occurring 2 to 3 times more often than declines. In contrast, technical events saw a major decline: throwing events were six times more likely to show a decrease in performance than an improvement, while jumping saw a 2.6 to 1 negative ratio.

Looking at the rankings in Figure 4, we see a linear divergence in performance evolution. While the top-ranked athletes improved by an average of 1.61%, the 35th ranked athlete declined by 1.51%. The transition to negative growth occurred at rank 22. Adult women were the notable exception to this trend, showing broad progress, with ranks 3 through 35 improving more than the top two positions. In contrast, the general decline was driven by male youth categories. U18 and U20 men exhibited the earliest drop-offs, with regression starting at ranks 6 and 9, respectively.

4. Discussion & Conclusion

The results demonstrate a systematic widening of the performance gap, particularly among men and young athletes. This increase in the performance range correlates directly with the 11.9% decline in German Athletics Association (DLV) membership and the massive 26.6% loss in the 27- to 60-year-old age group (Lei). This is the largest group of coaches, and their decline is particularly noticeable in technical disciplines, which rely on more intensive coaching.

Our analysis of the years during the pandemic also reveals a difference between disciplines. Running disciplines improved thanks to carbon technology (Willwacher et al., 2023) and uninterrupted training blocks (Venkatesan, 2022). Training facility closures led to a measurable drop in performance in the technical disciplines. Women, on the other

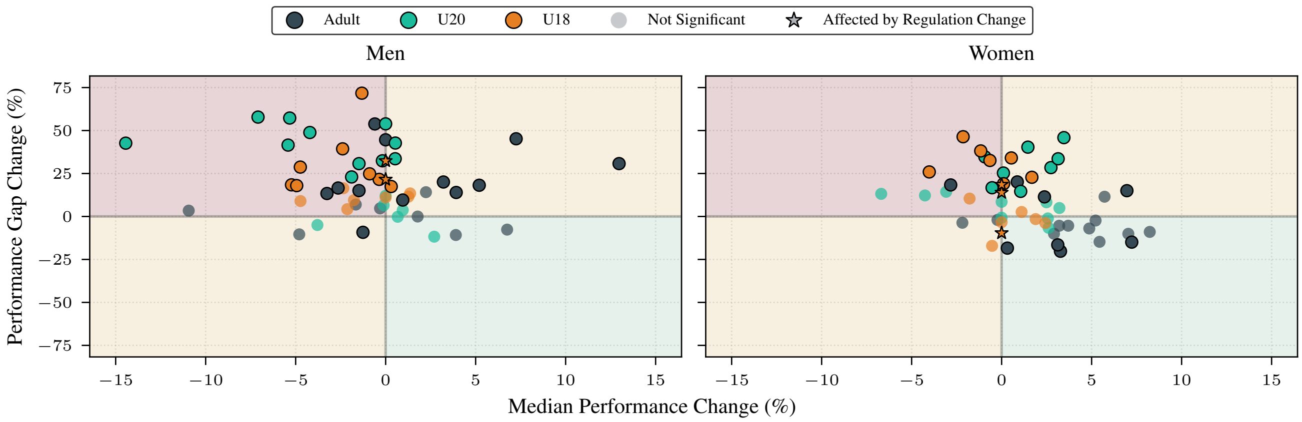

Figure 3. Change in median performance vs. performance gap (2001–2005 to 2021–2025, in %) for men and women across age groups. Marker transparency denotes statistical significance for performance gap (p ¡ 0.05). Median changes for disciplines with regulatory updates are fixed at zero due to restricted comparability.

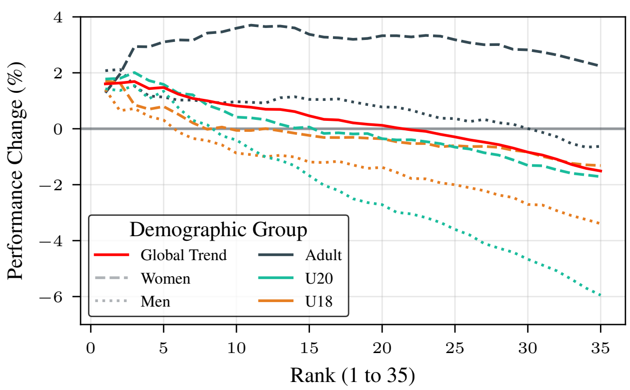

Figure 4. Average rank-specific performance development (2001–2005 to 2021-2025, in %), by age group and sex.

hand, show a catching-up effect, suggesting that professionalization is not yet complete.

The increasing polarization can also be attributed to the distribution of resources. National team athletes are secured through centralized funding. The extended elite relies on external support, which was truncated during the COVID-19 pandemic (Breuer et al., 2021). As a result, this extended elite has been lost. This lack of internal competition weakens international competitiveness in the long term, according to the adage ”Competition breeds excellence.”

The analysis is limited to the top 35 and thus represents the extended elite, not grassroots sports. This should be analyzed further in the future to understand the overall structural changes. The trend towards polarization explains the lack of medals at international championships. Future focus must be on strengthening the extended elite and coach training to revitalize competition, which will also benefit the top athletes.

Contribution Statement

The authors confirm their contributions to this work as follows: Maximilian Wernz performed data parsing, visualization, and Git repository management. Mattis Dietrich conducted the performance gap analysis. Luca Bouché performed the analysis of the impact of the COVID-19 pandemic. Eric Müller performed manual data cleaning. All authors conducted exploratory data analysis and contributed equally to the report.

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